

Markscheme

Mathematics: analysis and approaches
Practice question bank

169. (a)

A system of 3 linear equations in x, y, z row-reduces to the augmented matrix $\left(\begin{array}{ccc|c} 1 & 0 & 2 & 5 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 0 & m \end{array}\right)$.

Find the value of m for which the system has infinitely many solutions, and find one value of m for which the system has no solution.

the bottom row reads $0 = m$ (M1)

if $m = 0$ this is always true, giving infinitely many solutions **A1**

if $m \neq 0$ (e.g. $m = 1$) this is a contradiction, giving no solution **A1**

[5 marks]